Mixed-attack CICIoT2023 comparison

0.1 Mixed-attack CICIoT2023 comparison

Dataset configuration. We evaluate on a mixed CICIoT2023 stream constructed from 120,000 benign packets followed by six attack types at 50,000 packets each (DDoS-UDP Flood, DoS-SYN Flood, Dictionary Brute Force, MITM-ARP Spoofing, Mirai greeth flood, and Recon-OS Scan; 300,000 attack packets in total). Models train or calibrate on the initial benign prefix (`benignLimit`=60,000); all reported metrics are computed on the subsequent test region of 60,000 held-out benign packets and 300,000 attack packets.

Reading Precision and FPR. VAEESDD attains the lowest FPR (0.014) and the highest Precision (0.9966) in Table 1. These figures characterize a *conservative* detector: it raises few false alarms and therefore rarely mistakes benign traffic for attack. They do *not* imply superior detection overall. At this operating point VAEESDD recalls only 82.5% of attack packets (TPR 0.8251), well below Tenko (0.9801), and lags on F1, Macro F1, Accuracy, and AUC–ROC.

Matched false-alarm budget. In NIDS deployment, the operative question is detection rate at a fixed false-alarm budget rather than Precision at an unconstrained (often very conservative) threshold. Calibrating VAEESDD scores on the same held-out benign test mass to a matched $\sim$2% FPR yields TPR $\approx$ 0.8385, versus Tenko's TPR 0.9801 at FPR 0.0200. Thus the Precision/FPR advantage of VAEESDD disappears under a fair alarm budget: Tenko recovers substantially more attacks at the same benign FPR target.

Defensibility of the Tenko result. Under the reported operating point, Tenko leads on TPR, F1, Macro F1, Accuracy, and AUC–ROC while remaining at a practical low-FPR regime (FPR=0.02). Its AUC–ROC (0.9897) is also the best ranking among the compared methods, indicating that the advantage is not an artifact of a single threshold. Gains persist under a matched false-alarm budget, as above. The baselines are competent rather than strawmen: Kitsune is a strong streaming reference; Mateen is an adaptive deep baseline; VAEESDD is a frozen VAE recovered from a collapsed streaming configuration; and Isolation Forest follows Ramkumar et al., adapted here to AfterImage features for a fair feature-space comparison. ASP-based diagnosis is not applicable in this binary mixed-stream setting and is omitted.

Table 1: Mixed CICIoT2023 stream: binary detection metrics on the test region (60k benign + 300k attack). Tenko uses a benign-calibrated OR-rule operating point targeting approximately 2% benign FPR. Macro F1 follows the per-attack definition in `table_ix_mixed_with_iforest.csv`. VAEESDD is the frozen-after-benign variant (recovered from streaming collapse). Isolation Forest follows Ramkumar et al., adapted to AfterImage features.

Method	TPR	FPR	Prec.	F1	Macro F1	Acc.	AUC–ROC
Tenko[1]	0.9801	0.0200	0.9959	0.9879	0.9780	0.9801	0.9897
Kitsune	0.9637	0.1971	0.9607	0.9622	0.8758	0.9369	0.9733
Mateen	0.9048	0.0488	0.9893	0.9452	0.9189	0.9126	0.9665
VAEESDD	0.8251	0.0140	0.9966	0.9028	0.8861	0.8519	0.9540
Isolation Forest (Ramkumar et al.)	0.6115	0.0514	0.9835	0.7541	0.6769	0.6677	0.9347